DSA patterns → Kadane's algorithm

must be subarray contiguous → (sum / product)

$\boxed{\text{best}}$ → min / max

→ Sliding window will not work on negative values

* Flowchart for Kadane's algorithm

subarray / Array (negative, positive, zero)

↳ Kadane's algorithm

↓

Best → min / max

sum, product, frequency

→ if all elements are positive

example:- $\boxed{1 \mid 4 \mid 3 \mid 2}$ → subarray → sum → maxsum

* Template:

index by index

→ each index you need to take two choices

1. any answer ending at this index

2. nums = [1, 2, 3]
   0 1 2

1 | 2 | 3

1 | 1 2 | 1 2 3

→ ending at 3

↳ at each index i

↳ ending at that index

previous ans (or) arr[i] only

+ arr[i]

(i : 0 → n-1)

→ [0]
→ [1]
→ [2]
→ [3]
⋮
→ [n-1]

→ max → result

1 | 2 | 3

1 2 3 | 2 3

1 2

* example:-

nums = [-2, 1, 2, -4, 5]
0 1 2 3 4

ans : arr[0]
= ~~-2~~ ~~1~~ ~~3~~ 5

(i-1) → best ending

best ending = 1

best ending = 3

best ending = -1

best ending = 5

i = 0
↳ best = a[0] → -2

i = 1
↳ (-2+1) -1 | (1) 1

i = 2
1+2 → 3 | 2 → 2

i = 3
3-4 → -1 | -4

i = 4
-1+5 | 5

② Maximum sum subarray

nums = [-2, 1, -3, 4, -1, 2]
0 1 2 3 4 5

```
i = 0
best ending = a[0];
ans = [0];

for i in range(1,n):
    v1 = bestending + a[i]
    v2 = a[i];
    bestending = max(v1, v2)
    ans = max(ans, bestending)

return ans
```